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Basic Sources of Justification

Introduction

Before we get to the various ways of justifying basic premises, we need to make an important note: a premise having "basic justification" doesn't mean it's beyond doubt. Basic justification is always relative to some argumentative context. As an example, suppose your friend claims that Steve Jobs's decision to make *Siri* the flagship feature of the iPhone 4 was influenced by having seen the movie *Her* with Joaquin Phoenix, and you make the following argument against that claim:

> Premise 1: *People cannot be influenced by events that happen after their death.*
> Premise 2: Her *was released after Steve Jobs's death.*
> Conclusion: *None of Steve Jobs's decisions were influenced by the movie* Her.

Let's say your friend accepts premise 1, but thinks you're wrong about premise 2. You present the following argument to convince them:

> Premise 1: *Steve Jobs died in 2011.*
> Premise 2: *The movie* Her *was released in 2013.*
> Conclusion: *The movie* Her *was released after Steve Jobs's death.*

Now, if your friend doubts either of the premises in this last argument, it would be natural to look up the information on, say, Wikipedia. Wikipedia is a pretty reliable source for this type of information, and, if your friend is reasonable, they'll accept those premises once Wikipedia confirms them. Wikipedia can serve as a *basic source of justification* (BSJ) in this case. But this doesn't mean that information provided by Wikipedia is beyond all doubt. It simply means there are many contexts in which reasonable people will stop disputing premises, once they can verify those premises with a source like Wikipedia. It's the same with all sources of basic justification: they're not supposed to provide some profound metaphysical certainty, but rather a good enough starting point. They're what often establishes what we called *common ground* before: claims that everyone in a given context finds acceptable. And when they fail to provide such common ground, they're at least a way of shifting the *burden of proof*. It's now up to the other party to show that the justification doesn't work in this case.

Let's turn now to examining some BSJs and how they are able to justify various claims.

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BSJ1: Definitions and Logic

In some cases, we can tell a claim is true because its truth is a matter of *definition*. Consider the following two claims:

- *All bachelors are unmarried.*
- *All triangles have three sides.*

We know these claims are true simply due to the standard definitions of the words involved. A bachelor is an unmarried adult male; that's what the English word *bachelor* means. So if someone says (for whatever reason), "All bachelors are unmarried," you don't need to go out and check by interviewing a bunch of bachelors about their marital status. Similarly, you don't need to count the sides of a triangle to determine whether it's true that *All triangles have three sides*, because having three sides is part of the definition of the English word *triangle*.

Further, some claims are justified because they're *logical truths*: they're true simply because of the way logic connects ideas. Consider this claim: *Ling is either an accountant or she's not an accountant.* Whether Ling is an accountant or not is a question of fact that might be worth investigating. But the claim that she must be one or the other is logically guaranteed: there's no third possibility. Or consider this example: *If no As are Bs, then no Bs are As*. (Think about that one: for any example you can think of—that is, for anything you can substitute for *A* and *B*—if none of the *A*s are *B*s, then it must also be true that none of the *B*s are *A*s.)

Definitions and logic provide excellent basic justification. But for most claims—and certainly for many interesting and important claims—definitions and logic won't be enough.

BSJ2: Your Senses

In many cases, the justification for our beliefs is that we saw, heard, or felt it ourselves. Is it true there's a coffee cup on my desk? Yes, it is. I can see it with my own eyes. Did our competitor really offer me a job? Yes, I heard them say it! Is some form of odorous substance leaking from our production facility? Yes, I can smell it myself! Is such evidence reliable?

In some cases, it's nearly impossible for claims about our experiences to be false. If I think *It's dark here* because that's what I'm experiencing, or if I tell you *I ate lunch at Subway about an hour ago; I remember doing it*, those claims are very, very likely to be true. Some philosophers refer to such beliefs in terms of *entitlement*: you're immediately entitled to

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believe them. The point isn't that such claims couldn't be false—after all, it's possible that you're hallucinating! Rather, the point here is merely that such observations, and the claims that result from them, are so basic, so fundamental to how we conduct ourselves, that no one could blame you for believing them, even if you haven't made any special effort to verify them.

But our senses can let us down. Our eyes can deceive us if we're tired, or if it's dark or foggy. We can be mistaken about what we think we heard someone say when we're in noisy environments. And intoxication (whether due to alcohol, prescription drugs, or street drugs) can have a significant impact on how trustworthy our senses and memories are.

We should also doubt our senses when they tell us things that are contrary to what we have, on other grounds, good reason to believe. For example, when you see a stage magician "cut a woman in half," it isn't reasonable to believe that he really cut a woman in half. Everything you know about how the world works—including how the human body works—tells you that cutting someone in half and their living through the experience is impossible. If your senses tell you that's happening, you should doubt them, and admire the magician's clever illusion. Likewise, if you believe the robbers drove off in a black car, but four other reliable witnesses all claim it was a blue car that doesn't necessarily mean you are wrong, but it does mean you have reason for doubt.

Further, there are some situations in which we're simply not very good at describing just what we've seen. Imagine someone tells you "That car zoomed past here at over 100 km/h!" How reliable is that? Most civilians aren't especially good at judging the speed of a passing car. Can you really be sure the car was going 110 km/h, not 90 km/h?

We can roughly sort the reasons to doubt our senses into three categories. First, we can distinguish between reasons to think your senses may not have been working properly and reasons to believe that what you think you saw (or heard, or felt…) simply cannot be true. Let's call this second category *conflicting evidence*. The first category can be further divided by distinguishing between factors to do with you and factors to do with the environment. Let's call this first subcategory *personal factors* and the second *environmental factors*.

This gives us the following checklist we should go through before trusting our senses (or trusting somebody else who's relying on their senses).

1a) **Personal factors**: *I was drunk*, *I was tired*, *I wasn't wearing my glasses*, etc.
1b) **Environmental factors**: *It's dark*, *It's foggy*, *I was too far away*, etc.

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2) **Conflicting evidence**: *Other reliable people saw something different*, *What I seem to have seen conflicts with "common knowledge,"* etc.

Note that this list is just a guide to those factors that may give you reason to doubt your senses. It will usually not be necessary to determine the category into which such reasons fall (and sometimes the category to which a reason belongs will be unclear). What's important is to be able to tell when you're in the kind of situation where you have at least some reason for doubt.

BSJ3: Eyewitness Testimony

In some cases, the best evidence we have in support of a particular claim is that it was told to us by someone who has first-hand knowledge.

Imagine your friend Maggie says, "Martin was at Walmart this morning; I saw him there." Should you believe Martin was really at Walmart this morning (perhaps when he should've been at work)? The evidence we have is that Maggie says she saw him there with her own eyes. Is that enough? What reasons might we have to believe or to doubt Maggie? For most purposes, such testimony is pretty reliable. People generally know what they've seen, and they generally tell the truth about it. But 'generally' isn't the same as 'always'. It's possible, that Maggie is mistaken about having seen Martin. Her senses are subject to the same limitations discussed in the section above, including personal factors and environmental factors. Maybe she wasn't wearing her glasses or she was quite far away, and it was someone who merely looked a lot like Martin. It's also possible that she's not telling you the truth, although that would only be likely in special circumstances (circumstances in which you have reason to believe Maggie is lying, or bending the truth.)

One set of circumstances in which you have reason to doubt people's truthfulness is important in business, and so deserves special mention here. People are more likely to tell you a lie, or omit some part of the truth, when they stand to gain something from doing so. Imagine you're thinking about buying new speakers for your car and your friend says to you "I have this new speaker system in my car and you can really hear the difference this new technology makes; it's an entirely new and much improved listening experience." Generally, you'd be inclined to think your friend is telling you the truth. Now imagine hearing this from a salesperson at a store that sells that speaker system. You'd be justifiably suspicious. The salesperson has a *vested interest*: they stand to gain, financially, if you accept their advice and buy the speakers. Note that a person with such a vested interest might still be telling the truth. So you shouldn't conclude that the speaker system must be bad. It's just that it's usually better to avoid relying solely on the report of someone with a vested interest. It makes sense to seek some independent confirmation.

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How Reliable Is Eyewitness Testimony in Court?
Interview with Dr. John Turtle*

1) What, in general, does the expert literature say about the reliability of eyewitness reports?

There is no overall measure of eyewitness accuracy or inaccuracy. Nor is anyone trying to develop one; but research shows that people's memory for events they see for relatively short periods of time is susceptible to a number of factors that limit its accuracy and completeness. One way to gauge the effect of these limitations is to analyze cases in which people have been exonerated of crimes they were thought to have committed, usually on the basis of DNA testing of evidence retained from a crime scene. Out of hundreds of such cases of error reviewed over decades of research, mistaken eyewitness identification was involved about 70% of the time—more than any other factor, by far. So we know people make mistakes, and that criminal investigation procedures and the court system often don't catch these mistakes. Knowing that eyewitness evidence is so prone to error, and that it can be overly relied upon by judges and jurors, has led some to say that it should never be used in criminal investigations and trials. The reality, however, is that sometimes it's the only evidence available to get an investigation started, or it's important corroborating evidence, but most eyewitness researchers would recommend it not be used as the sole or main evidence in a case.

2) What are some of the main factors thought to account for the low reliability of eyewitness reports?

A helpful way to organize the factors that affect eyewitness memory is to think of two categories: *estimator variables* and *system variables*. People new to the eyewitness memory area often focus on estimator variables at first—factors whose effect on memory we can only estimate, such as exposure time, viewing conditions, stress, age, retention interval, and others. Estimator variables are especially difficult to work with, because something like exposure time is often estimated to begin with (e.g., "I *think* I saw the guy for about 10 seconds."), plus its effect on memory can also only be estimated because so many other factors are involved that likely interact with each other. So, we assume that overall, more exposure time is likely better than less, shorter retention intervals are likely better than longer ones, more light is probably better than less light. Knowing your memory for an event will be tested later probably improves memory performance, but nobody can answer the question of whether or not a particular fifteen-year-old female witness who saw a forty-year-old male stranger for fifteen seconds at night from six metres away is likely to be accurate in her description or identification of the offender three weeks after the event.

The other category of factors that affect eyewitness memory is system variables—factors over which the legal system has at least some degree of control, such as

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how witnesses will be interviewed, how photos for a police lineup will be selected, how those photos will be presented, who conducts the lineup, and how the information will be used in an investigation or trial. For example, a witness's memory can be distorted if an interviewer includes in a question information about something the witness did not see; an investigator who knows which photo in a lineup is the suspect's can inadvertently suggest to a witness that they are "right" or "wrong"—and a witness who is told they are "right" after their lineup decision can come to have more confidence in their choice than someone who is not given that information.

*Dr. John Turtle is an Associate Professor in Toronto Metropolitan University's Department of Psychology, and an acknowledged expert on eyewitness testimony.

BSJ4: Common Sense

One key source of information—one we use every day—is our own, admittedly loose, understanding of how the world works. We use this built-in understanding constantly; we need to simply to get through the day. We know without thinking, for example, that if we tip a cup of water over, the water will flow all over the desk. We know, without thinking it through, that if we put a pot of water on the stove, and turn the stove to *high*, the water will get hot. And we understand, without having to be told, that if we criticize an acquaintance's taste in clothes, we'll hurt their feelings. All of this is automatic: we don't need to gather detailed information about *this* cup of water or *this* stove, or *this* acquaintance. We understand fairly well how things and people work, and so we're able to understand and even predict their behaviours.

This intuitive understanding of the world has complex origins. Part of it may be biological. There's evidence, for example, that even newborn babies have some understanding of the fact that solid objects can't pass through each other, and that an object hidden from view has not disappeared but is simply unseen. Babies don't need to learn this stuff by exploring the world themselves. They are apparently born with that knowledge. Babies are also apparently born knowing that faces belong to a special category of things with special meaning in the world. Babies are also born with at least some innate understanding of numbers, even though they don't have the words to express it.

Other elements of this common sense have to be learned. We have to learn, sometimes through painful experience, that fire will burn us; that's not something we're born knowing. We have to learn how various mechanical devices (like doorknobs) work. And we have to learn many aspects of how our fellow human beings think and react to various circumstances. But the net result is that, due in part to inherited capacities and in part to learning, we all have an intuitive understanding of crucial elements of our world, and this understanding helps us navigate the world in countless important ways.

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The three examples above (the cup, the stove, the acquaintance) were chosen with three important categories of common sense in mind: elementary physics, simple design knowledge, and folk psychology.

Elementary physics is the knowledge of how physical objects generally behave here on earth. Here are a handful of claims that belong to our shared sense of elementary physics:

- *Objects fall to the ground if not held up by anything*.
- *Fire burns many things*.
- *Water runs downhill*.
- *Steel is stronger than glass*.
- *People cannot pass through solid objects*.

Elementary physics is incredibly important in getting through the day. It helps us make simple, yet important, decisions such as not leaving a building by walking out the window on the fifteenth floor, not making a spare key made of wax, and so on.

Note that knowing elementary physics doesn't imply that you could pass as much as a grade six physics test. Knowing that water flows downhill doesn't mean we know the relevant theories and equations. We all know these elementary facts about the physical world, even though we might be utterly mistaken about the best current theories about what lies behind these simple claims. For example, everyone knows that if you drop a feather and a heavy suitcase off the top of a tower, the suitcase will fall to the ground faster. But many people think this is because heavy objects generally fall faster than lighter ones. This was proven false over four hundred years ago. The real answer has to do with air resistance, but that part isn't intuitive.

Similarly, we have a shared sense of how certain man-made objects and structures generally function. If you get into a car you've never driven before, how do you know that pressing the pedal on the right will make the car go faster? Because that's how cars are designed—the gas pedal is on the right. Everybody knows that. When you pick up a hammer, you know it'll be effective at driving nails because you know that's what hammers are designed to do. And if you're visiting a friend you will probably need no special instructions for how to flush the toilet or how to use their toaster.

It's worth noting that this common-sense design approach can work even with regard to our understanding of organizations. When you arrive at the front desk of a law firm, you expect to be greeted by a secretary or receptionist, because you know that that's how law

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firms work. If you're unhappy with the service at a restaurant, you may ask to see the manager, knowing that there's likely to be such a person, and that they're likely to try to accommodate you to some degree. Almost all restaurants are designed to operate that way.

As these examples make clear, design knowledge is tied to having grown up in a particular culture. (If you grew up in Australia, for instance, it would be obvious to you that you're supposed to drive on the left, rather than the right.) Thus, simple design knowledge can serve as a basic source of justification only where everyone involved has strong familiarity with the cultural context in question. When travelling to places with significantly different cultural norms, for example, it's not advisable to rely on your own expectations of how things are designed or organized.

Finally, there's *folk psychology*,[1] the body of knowledge we all have that allows us to correctly attribute to other people (and sometimes animals) moods, beliefs, desires, intentions, memories, and other psychological phenomena and experiences. This body of knowledge is useful because it allows us to make use of the fact that other people have such psychological states, which enables us to explain and predict their behaviour. This is an invaluable skill in interacting with others, whether in a bar, in an office, or when negotiating a deal.

Our capacity for folk psychology is truly essential; it's what allows us to live and work together in groups. Folk psychology is what lets us know that punching someone will make them mad, that gossiping about them will hurt their feelings, and that flirting with their romantic partner might make them jealous. We know intuitively how their minds work in such situations. And we can often predict how they might act in response. Folk psychology is also what allows us to motivate them to alter their behaviour: we know that they'll tend to do more of something if we reward them and less if we punish them for it (although this isn't always true).

Our capacity for folk psychology is, however, imperfect. In some cases, there may be cultural differences that make it hard to interpret someone's behaviour (*Huh? Why did he bow? What does that mean?*). More generally, sometimes we just don't know what's on someone else's mind. Sometimes people make no sense to us: we see their behaviour, but can't map it onto any particular understanding of what's on their mind. Sometimes that's the result of lack of imagination on our part. Other times, it may simply reflect the fact that people are different—we don't all think identically. So we may find ourselves truly puzzled and ask ourselves *What on earth are they* doing*?* But the very fact we sometimes ask this

[1] The term *folk psychology* was coined by philosopher Daniel Dennett in 1981, and discussed in his 2013 book, *Intuition Pumps and Other Tools for Thinking*.

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question reinforces the fact that we understand other people as creatures with minds, beings that have beliefs and act upon them. If we didn't see other people that way, that is as doing things intentionally, we wouldn't bother to wonder why they do what they do.

Sometimes we're unable to predict a particular person in a particular situation because we know that the person is subject to different motives at the same time. Consider the ultimatum game, in which one player is asked to divide an amount of money between herself and the other player. The other player gets to decide whether to take the deal or reject it (in which case the "dealer" keeps all the cash). If the first player offers a very uneven split, it's hard to predict what the second player will decide because we know she's subject to two different motives. On one hand, she's probably like most people in that she'd prefer some money to no money. On the other hand, again like most people, she probably wouldn't want to accept a deal she sees as unfair. Different people will balance those competing values in different ways, which makes the outcome harder for us to predict.

This last example has important implications in a business context. One of the dangers of studying some parts of the typical business-school curriculum is that it makes people forget or ignore some essential parts of folk psychology. For example, everyone knows that people are sometimes motivated by a sense of loyalty. This means that cultivating a sense of loyalty among employees is an important tool for corporate leaders. However, since loyalty is hard to quantify, it's been underemphasized in comparison to monetary rewards by large parts of economic theory (including the decision-theoretic models discussed in this course). This carries the danger that leaders schooled in economics will focus on external rewards and punishments when devising incentive schemes, neglecting potentially more effective intrinsic motivating factors like loyalty.

These three forms of knowledge—our knowledge of elementary physics, simple design knowledge, and folk psychology—together constitute the important basic source of justification we call *common sense*.

You may have noticed that what we've called *common sense* here is closely related to system 1 thinking as described in this course's introduction. As we said there, such thinking is absolutely essential for living daily life, and remarkably successful in many contexts. However, it's also subject to systematic limitations and errors. For now, we need to urge caution: while we can often depend on common sense as a basic source of justification, it can also be illusory.

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BSJ5: Expert Testimony

Common sense is strongest in situations that are familiar to everyone. When it comes to unusual situations, or unusually obscure topics, common sense regularly misfires, and it's in these contexts that we might want to turn to experts.

An *expert* is someone who has great knowledge and experience in a particular field, and who thus has greater access to relevant facts and an enhanced ability to make judgements about them than others. Since many people claim to be experts, it's not always easy to identify those who truly are. The following are some common markers that lend initial plausibility to the claim that somebody is an expert:

- Prolonged study or advanced degrees (such as PhD)
- Extensive experience in dealing with a particular topic
- Recognition by other knowledgeable people in their field

But these markers aren't entirely reliable. Somebody may have gained a PhD, for instance, by spending a lot of time working on a theory that's been thoroughly disproved in the meantime; they may have extensive experience but a rather spotty record at it, or they may work in a field that's been discredited as a whole (such as homeopathy). So when we ask whether we can rely on an expert, we should be looking, in addition to the markers mentioned above, for a track record of providing accurate information and good advice—or at least for the absence of a track record of providing inaccurate information and bad advice.

A further note of caution concerns the fact that expertise is specific to a particular field, and often quite narrow. Being an expert on one topic doesn't make you an expert on another topic. Most people would agree that, of course, the opinions of celebrated experts in sports, science, or entertainment should carry no additional weight in questions outside their respective domains. What's less well known is that even medical doctors are typically highly unreliable in diagnosing illnesses outside their area of specialization.

It's also important to note that there are many fields in which experts disagree about important questions. In these cases, we can't simply rely on the opinion of one expert. In addition, we need to be on our guard, for an expert might not bother to mention that some of their views are controversial. But even when experts disagree, their disagreements can be informative. First, an expert in a field can teach us constructive ways of thinking about a field, even if they can't give us settled answers. Second, expert disagreement often occurs against a backdrop of large areas of agreement. And when we see experts agree on issues

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A, *B*, and *C*, but disagree on *D*, that reassures us about *A*, *B*, and *C*, while also helping us see where the complexities and uncertainties lie.

"I read it on the Internet!"
One of the most accessible and powerful sources of information available to us is the Internet. And while sources of information on the Internet vary enormously in terms of reliability, we all rely so heavily on the Internet that it deserves special mention.

The Internet, of course, isn't a single source of information. It's a tool that makes available to us everything from newspaper stories to corporate press releases to government policy statements to gossip posted on social media to conspiracy theories to deliberately false stories and, increasingly, to AI generated summaries of all of those.

As a result, we need to look carefully at the very different levels of reliability of the different parts of the Internet. Imagine you are looking for information about the relationship between smoking and lung cancer. The following sites might all have *something* to say about that:

- The website of the World Health Organization
- The website of the Heart and Stroke Foundation
- A blog written by someone with lung cancer The website of a cigarette company
- A research article published on the website of the *British Medical Journal*
- The website of an electronic cigarette (or "smokeless" cigarette) company
- A story about lung cancer on the *New York Times*'s website
- A friend's Facebook posting about what she thinks really causes lung cancer
- A blog written by a cancer researcher

Clearly, these sources of information won't all be equally reliable. Some of them are highly likely to put forward only claims supported by expert opinion and substantial data, with no interest in doing anything but providing you with reliable facts. Some of them, on the other hand, will be trying to convince you of things that are in *their* interest to have you believe (perhaps because they want to sell you something). Others may be well-intentioned, but simply lack the expertise required in order to reliably judge the existing evidence. So, make sure to exercise caution.

BSJ6: Scientific Studies

One important way we can call on experts is by relying on their published work, especially peer-reviewed studies. A study (or any piece of academic writing) is peer-reviewed when approved, before (or in some cases after) publication, by other experts in the field. Thus, if a medical researcher has been conducting experiments with a new drug and wants to publish the results in a respectable medical journal, the journal will ask other medical researchers (the original researcher's peers) to check whether the study or article conforms

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to the highest scientific standards. Only after these reviewers have given the go-ahead can the study or article be published. This is generally regarded as the gold-standard of scientific publishing (though there are some academic fields, like law, that have developed different models).

Now, it's quite common for people to ask you to believe a claim because *studies show* or *they did a study on that, and found*…. Scientific studies are often an excellent source of justification, especially when we don't just rely on one study but on a variety of studies on the same topic, all of which confirm our claim. However, in most cases the path from scientific studies to the premises of an argument isn't straightforward. And it's common for a lot of credibility and accuracy to be lost along the way. We can't get into detail about how to evaluate scientific claims here, but the following is a very useful rough-and-ready guide.

1) You can usually disregard any claim about what *studies show* if the person making that claim is incapable of citing a specific, credible source.
There are a couple of reasons for this. First, people sometimes say *studies show* just to fake expertise when they have none. Second, for almost any claim—no matter how far from the truth—there's likely to be some kind of study somewhere that supports it. So if a study is out there, it's important that we're able to check that the study was conducted in accordance with the highest scientific standards. And we can't do that if we're not given a source. (Note that this doesn't mean the claim in question has to be false; it just means that *studies show* is not a source of justification in these circumstances.)

2) Be careful as to whether the study being cited actually supports the claim being made or whether it instead supports a slightly different claim.
The problem is that scientific studies are often misreported in small and big ways. Most people don't get their information about scientific progress directly from scientific journals—they get it from social media or the popular press instead. And those sources, even respectable news outlets, tend to use headlines geared more towards generating interest from readers than accurately representing scientific results. Thus, it's important to read past the headline and see what the researchers actually found. If the headline says *Researchers Find Cure for Cancer*, you may well find that what the study actually found is a small clue that could someday lead to a partial cure for some forms of cancer.

3) Be careful when a study goes against established scientific consensus.
Of course, sometimes the consensus is mistaken, and a new study is important precisely because it shows this is the case. But most of the time, when one or a handful of studies are on one side of the issue while the overwhelming majority support the other side, it's

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the majority of researchers who have it right. Here, again, is a reason not to blindly trust popular science reporting. Social media and the popular press tends to over-report new, surprising, and revisionary results, because those generate exciting headlines. When it turns out, at some later time, that these exciting results don't hold water after all, this is much less likely to be widely reported or reported at all.

In short, when confronted with the *studies show* form of justification, you should (a) ask for the source, (b) check the source for credibility and accuracy, and (c) check whether the results are widely accepted in scientific circles.

Wikipedia

Wikipedia is a free online encyclopedia. The greatest thing about Wikipedia is that anybody can contribute. If you know something about French cooking, you can add information and insight to the many pages on that topic. If you have special knowledge about your favourite sports team, you can edit that page and share your knowledge with the world. Literally millions of people have contributed their knowledge to Wikipedia, making it a truly astonishing source of information.

But Wikipedia's greatest strength is also its greatest weakness. The fact that literally anybody can edit most Wikipedia pages means that even people who aren't knowledgeable can make edits. And some people use Wikipedia to promote their own strange views or personal interests. The good news is that there are many volunteers constantly watching their favourite Wikipedia pages and double-checking each other's work, which means that false or biased information is typically corrected quite quickly. Some studies by fact-checkers have suggested that, on average, the reliability of Wikipedia is even higher than the reliability of printed encyclopedias written and edited exclusively by certified experts.

However, some Wikipedia pages are more reliable than others. In part, this reflects the fact that some pages simply haven't been worked on very much by knowledgeable individuals. But it's also the result of the fact that some topics are more controversial than others. And when topics are controversial, people on different sides may try to lend their own views greater credibility by presenting them as incontrovertible facts on Wikipedia.

For example, pages about topics related to current politics, or long-standing ethical or military disputes, can be so controversial that people with strong opinions have been known to use the relevant Wikipedia pages as a place to make their own views known, or to advance causes they care about. For instance, if you want information on questions like *Is Crimea part of Russia? Is abortion wrong?* or *Whose fault is the dispute between Israelis and Palestinians?* you should be cautious about what you read on Wikipedia.

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The pages on basic physics, biology, geography, history, and mathematics, on the other hand, aren't terribly controversial, and there's little worry about bias or ideology interfering. So, if you want to find out how many bones are in the human body, or the population of Latvia, or what year the US passed the *Foreign Corrupt Practices Act*, Wikipedia is a pretty good source.

Justifying Normative Claims

For any claim to serve as an argument's premise, it has to be acceptable to the person to whom the argument is addressed. This is also true for normative claims including those that are intended to function as bridge principles. As with descriptive claims, one way to ensure this is to first give an argument that has the bridge principle as a conclusion. Since bridge principles are themselves a type of normative claim, such an argument would have to have (at least some) normative premises. And, again, just as with descriptive claims, at some point such argumentation comes to an end and you simply have to accept some basic premises. Unlike in the case of descriptive claims, however, there are no easily agreed upon ways in which we can come to know basic normative claims. We can't literally see, smell, or taste whether torturing innocent people is wrong, for example. We either agree to it as a starting point or we don't.

Thus, when there are disagreements about very basic normative claims, it's not easy to see how to resolve them. However, we often don't need to go back to the most basic forms of justification. More often than not, we can simply rely on the fact that the person we're addressing likely accepts our basic premises. The same is true for broad, general, normative claims like many useful bridge principles. Most people, for example, would not ask you to justify the claim that it's wrong to torture innocent people. For most purposes, such a premise can simply be put forward without any special effort to justify it. Many bridge principles will be accepted by the audience simply because they express shared moral or ethical principles (like *stealing is wrong*); others will be accepted because they express a shared purpose (such as *we should only expand into the European market, if it is profitable*).

This doesn't mean you'll never be asked to provide an argument for a bridge principle (or any other normative claim) you're using. Take one of the examples above: *We should do whatever maximizes our profits*. While this claim is often heard, you can be sure to encounter resistance to it in many contexts. If you want to use it, you need to be prepared to provide an argument for it.

Finally, note that bridge principles are often left implicit, so they take the role of hidden premises in many arguments. For example:

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Premise: *Andruw is the most qualified candidate*.
Conclusion: *Andruw should get the job*.

This is an argument that apparently moves directly from a descriptive premise to a normative conclusion. Thus, it can't quite work exactly as stated here. What's missing is a bridge principle. In this case:

Premise 2: *The most qualified candidate should get the job*.

This is an example where leaving the bridge principle unstated in the original argument isn't a big deal, because most people would accept it anyway. But other cases won't be so easy. Consider this argument:

Premise: *Reducing carbon emissions would require a change to our lifestyle*.
Conclusion: *We shouldn't reduce carbon emissions*.

Again, we have a descriptive premise and a normative conclusion. What's the bridge principle that would work to connect the two? Probably this one:

Premise 2: *We shouldn't do anything that requires a change of lifestyle*.

Some people would probably accept that particular bridge principle, but many others wouldn't. Thus, leaving it as an unstated or hidden premise is problematic. The argument leaves out an important part that's likely to be challenged if made explicit.

Conclusion

We've laid the foundation for the identification and evaluation of premises. We've seen how beliefs and claims are related, and that the best standard to use to evaluate them is the justification of belief and the acceptability of claims. We've also briefly discussed various ways in which claims can meet the standard of acceptability.